Experiments on the Postbuckling Behavior of Circular Cylindrical Shells under Torsion

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1. Introduction

It is one of the basic problems on the stability of structures, to clarify the buckling and postbuckling behaviors of circular cylindrical shells under torsion. Hence, numerous researches have been made on these subjects. Concerning the buckling problem, approximate solutions were obtained by Donnell [1] and Batdorf [2] under two different boundary conditions, which were ascertained to be in fair agreement with experimental results [1, 3]. Later, based on both Donnell and Flügge equations, accurate solutions under eight sets of boundary conditions were obtained by the author [4] for a wide range of shell geometries. Hence, the buckling problem seems to be almost fully investigated.

As for the postbuckling problem, solutions of rough approximate nature were obtained by Loo [5], Nash [6] and Hayashi [7], and the corresponding experimental researches were performed by Nash [8] and Weingarten [9]. However, it seems that the agreement between theory and experiment has not been achieved. On the other hand, the so-called initial postbuckling problem was solved by Budiansky [10] to estimate the initial imperfection sensitivity.

Recently, the author conducted precise experimental studies on the postbuckling behavior of clamped cylinders under external pressure [11] as well as edge compression [12], the results of which were confirmed to be in reasonable agreement with the corresponding theoretical results [13, 14]. As a continuation of these studies, the present research aims to clarify experimentally the whole aspect of the postbuckling behaviour of cylindrical shells under torsion.

Test cylinders and test apparatus are almost the same as those reported previously except for the loading system, for which great care is taken that the cylinder is subjected to purely torsional loads without the effect of bending and axial forces. Although a slight axial constraint is observed to remain, the critical load and the corresponding wave number for each test cylinder are found to agree well with those theore-

tically predicted [4], and the present results are considered to provide fundamental data for elaborate theoretical analyses in the future.

2. Test Specimen and Test Apparatus

As stated in the previous paper [11], test cylinders with radius $R = 100$ mm and thickness $h = 0.247$ mm are made of polyester film with Young's modulus $E = 5.55 \cdot 10^3$ MPa ($= 567$ kg/mm^2) and Poisson's ratio $\nu = 0.30$, by lapjointing along the longitudinal seam and attaching duralumin cover plates along the ends. Only the thickness of the cover plate was changed from 11 mm to 18 mm. The lengths L of the cylinder are chosen so that nominal values of the geometric parameter $Z = \sqrt{1 - \nu^2} L^2/Rh$ are 20, 50, 100, 200, 500 and 1000. Actual dimensions of test specimens are listed in Table 1. Preliminary experiments are carried out using the previous specimens, with almost the same results for the postbuckling behavior as here presented.

Table 1. Dimensions of test specimens
($R = 100$ mm, $h = 0.247$ mm; $E = 5.55 \cdot 10^3$ MPa, $\nu = 0.3$)

No.	20-5	50-4	100-4	200-4	500-4	1000-4
L [mm]	22.9	35.9	51.0	71.2	113.8	160.8
Z	20.2	49.8	100.3	195.8	500.2	998.6

A schematic diagram of the test setup is shown in Fig. 1. In the figure, *1* is the table of a tension test machine, *2* is a base plate, *3* is a test cylinder and *4* is a stiffening steel plate attached to the cover plate, to which four equidistant cylindrical rods *5* are fixed. To counter-balance the weight acting on the cylinder (about 6.1 kg = 60 N), two sets of the weight *6* are hung with a thin wire *7* through a pulley *8*.

The loading system comprises from *9* to *23*, in which *9* is the crosshead of the test machine, *10* is a reduction gear box and *11* is the lever to rotate the shaft *12*. To avoid the effect of eccentricity and axial constraint, an universal joint *13* and spline shaft *14* are used, to which a torque meter *15* with capacity 200 Nm is connected. The loading head *16* has a circular disc with four equally spaced holes and the twisting moment is transmitted to the test cylinder through the ball bushing *17* and the rod *5*. The loading head is held in the axial direction by a ball bushing *18* in the casing *19*, which is fixed to the stand *20*. A cylindrical cap *21* is assembled to the loading head through two sets of thrust bearings and by rotating it with the lever *22*, the relative position of the loading head to the test cylinder can be adjusted. With this system stated in the foregoing, almost pure torque can be applied

to the cylinder without significant effects of bending moments and axil forces. Combined loading of torsion and compression is also possible by attaching a load cell *23* to the lower end of the loading head.

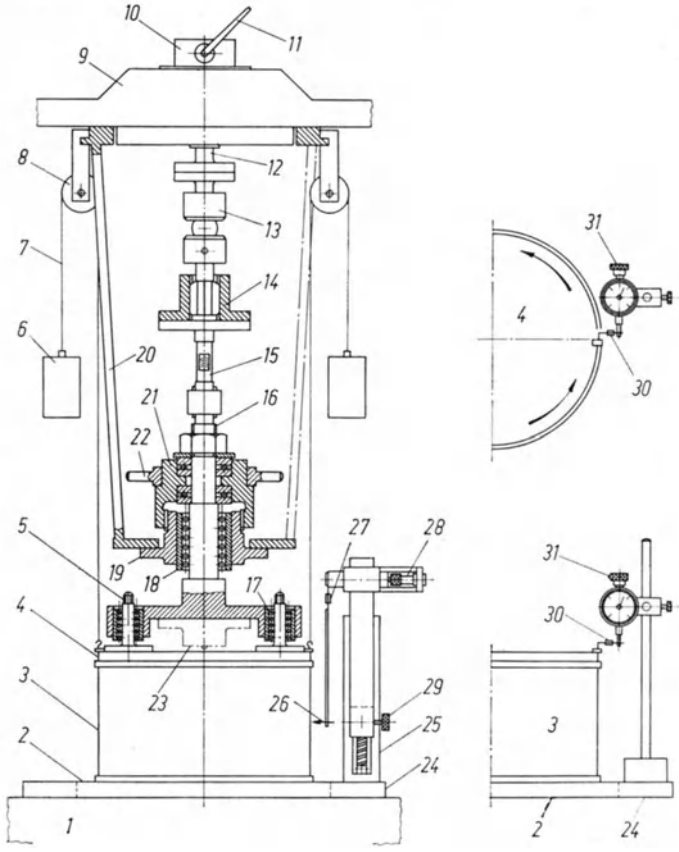


Fig. 1. Schematic diagram of the test setup.

The deflection measuring apparatus, comprising from *24* to *29*, is the same as stated in the previous paper [11], with which both axial and circumferential distributions of the deflection can be measured precisely without affecting the buckled surface. The axial shortening of the cylinder is measured utilizing the strain gauges *30* bonded to a thin lever fixed at the end of a dial gauge *31*, which, in turn, is used for calibration. Two sets of these are used in series to obtain the average value of edge shortenings at diametrically opposite points on the stiffening plate. The same sets are used for measuring the angle of twist by arrang-

ing them as shown in the upper right figure. A six-element dynamic strain meter is used for strain measurements while a X-Y recorder is used for recording the results.

3. Typical Test Results

To save the space, only the results for the test specimen No. 500-4 will be stated in detail. The relation between the torque T and the angle of twist ψ is shown in Fig. 2, in which the solid curve $0 - a - b - c - d - e - 0$ represents the actual result. The point a is the critical point, curves ab and de correspond to the unstable dynamic snap-through, while bcd to the stable equilibrium path. The snap-through occurs due to a relatively low torsional rigidity of the loading system, which is estimated to be about $1.9 \cdot 10^3$ Nm/rad, while the hysteresis loop in the range bc indicates a slight friction to the axial movement of the cylinder. (With the test specimen No. 1000-4, experiments are carried out by changing the axial load stepwisely, from which the frictional force is estimated to be less than 1 kg = 10 N, which corresponds to about 1% of the critical compressive load.) Assuming that the loading system is stiff enough and the frictional force is constant, we will obtain the curves depicted by dotted lines, from which the result for the ideal case with neither friction nor snap-through can be estimated as shown by the chain line. It will be seen that the equilibrium torque after buckling decreases monotonously with twist in the present range of experiments.

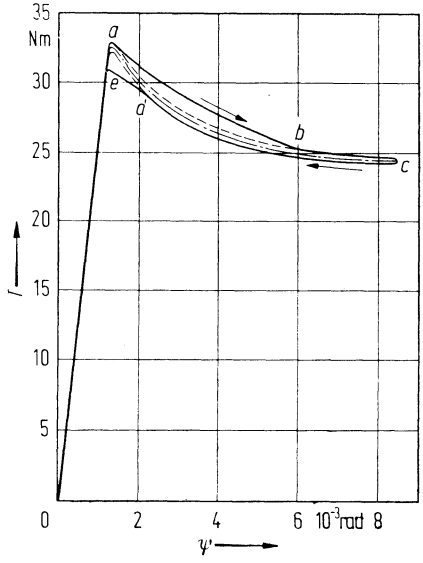


Fig. 2. Postbuckling relation between torque T and angle of twist ψ (specimen No. 500-4).

The relation between the torque T and the axial shortening δ is given in Fig. 3. As before, the chain line represents the result estimated for the idal case. It will be seen that after buckling, the length of the cylinder decreases monotonously with the development of buckled waves.

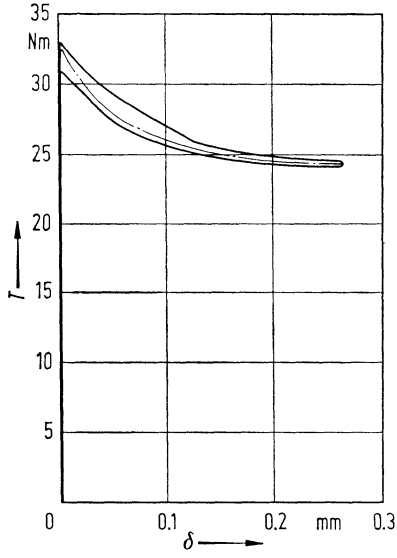


Fig. 3. Postbuckling relation between torque T and axial shortening δ (specimen No. 500-4).

To see the overall distribution of buckled waves, circumferential distributions of the deflection w (positive inward) are recorded along various circular sections, at a typical postbuckling configuration with $T = 25.1\,\mathrm{Nm}$ and $\psi = 5.43 \cdot 10^{-3}\,\mathrm{rad}$. The result is shown in Fig. 4, in which x and θ stand for the axial and angular coordinates, respectively. It will be seen that the number of buckled waves n is 15 and that the wave distribution is quite regular irrespective of the longitudinal seam located at the right side blank portion.

In the meanwhile, axial distributions of w are recorded along various angular positions, a part of which extending about one wavelength is shown in Fig. 5. The contour lines for the buckled surface are obtained from these with the results shown in Fig. 6. In the figure, solid and dotted lines correspond to the inward (positive) and outward (negative) deflections, respectively. Axial distributions of w passing through the valley and crest along the central section are also shown. It is to be noted that the waveform is symmetric with respect to the center of each buckle and that the buckled surface is almost covered with inward buckles as the angle of twist increases. It is to be added that the num-

ber of buckled waves n remains unchanged in the range of experiments here conducted.

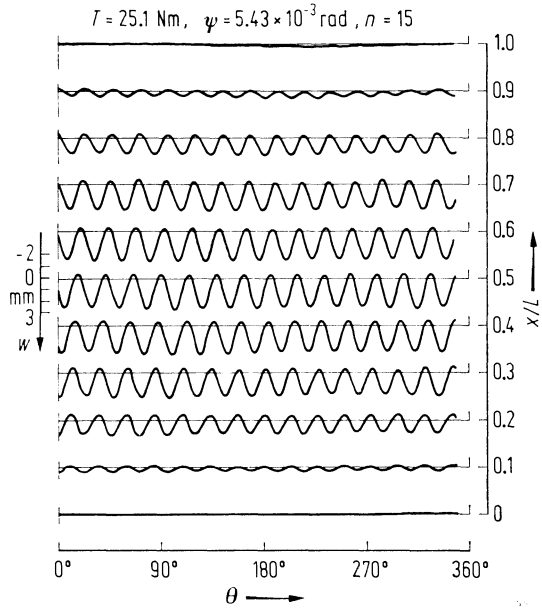


Fig. 4. Circumferential distributions of the deflection at various circular sections (specimen No. 500-4).

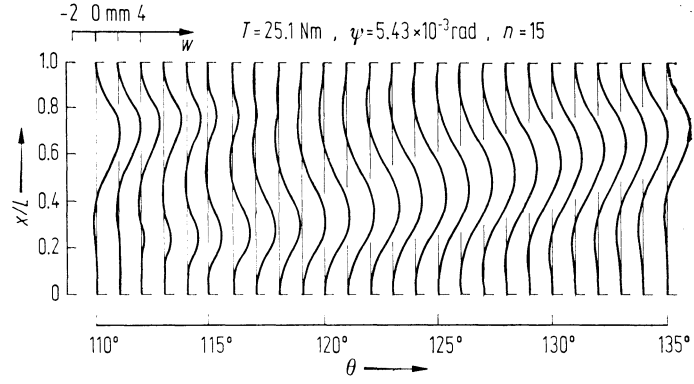


Fig. 5. Axial distributions of the deflection at various angular positions (specimen No. 500-4).

Finally, variations of the maximum inward and outward deflections along the central section of the shell, w_m, with applied torque T, are obtained with the results shown in Fig. 7. The chain line corresponds to

the ideal case as before. It will be seen that with the increase in twist after buckling, the inward deflection increases monotonously while the outward deflection soon attains its maximum value and then decreases.

Similar experiments are carried out for each test specimen. It is found that the number of buckled waves remains unchanged during the

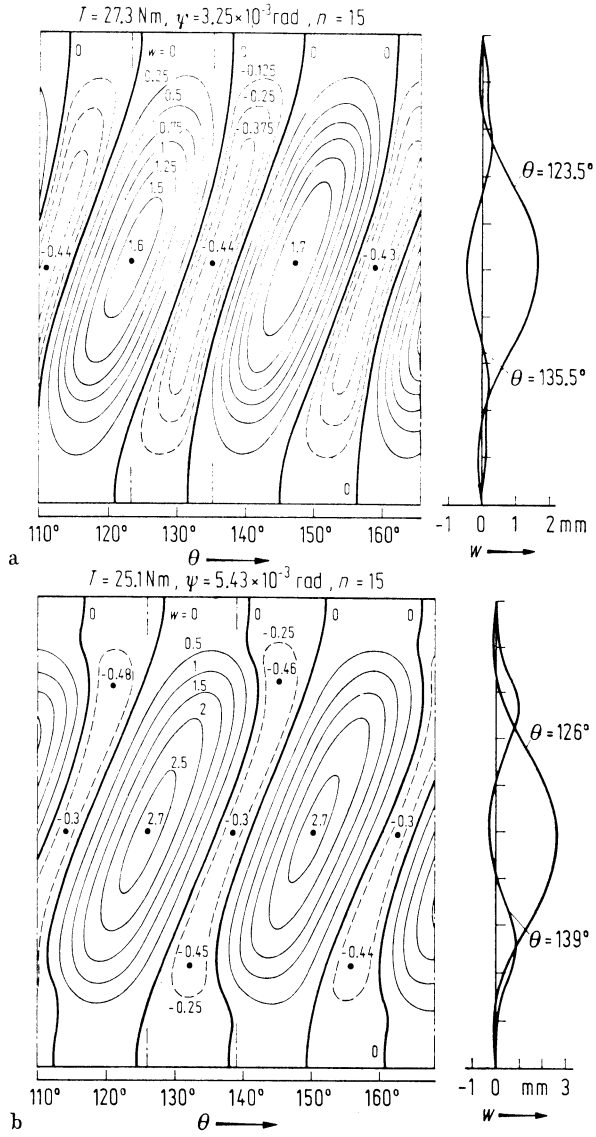


Fig. 6 a and b. Contour lines for typical postbuckling configurations (in mm, specimen No. 500-4).

test, although for relatively long shells with $Z \geqq 200$, the equilibrium states with the smaller wave numbers can be realized artificially in the fully buckled region. In general, the test can be repeated with almost the same result, although slight reductions in the buckling and postbuckling loads are observed in the immediate successive tests.

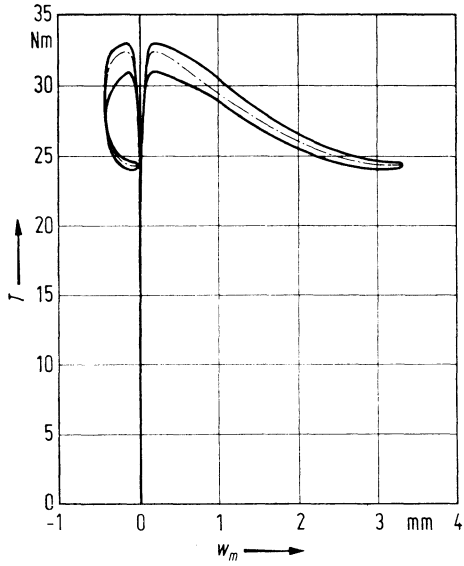


Fig. 7. Postbuckling relation between torque T and maximum central deflections w_m (specimen No. 500-4).

4. Buckling Load and Wave Number

To examine the degree of accuracy of the present experiments, critical loads and the corresponding wave numbers here obtained are compared with those theoretically predicted under the completely clamped boundary conditions, $u = v = w = w_{,x} = 0$ [4]. The results are shown in Fig. 8, in which k_s and β are the nondimensional load and wave number factors, respectively, defined by

$$k_s = \frac{TL^2}{2\pi^3 R^2 D} = \frac{\tau h L^2}{\pi^2 D}, \quad \beta = \frac{nL}{\pi R}, \quad D = \frac{Eh^3}{12(1-\nu^2)}. \tag{1}$$

Some of the previous experimental results [1, 3] are also shown for reference. It will be seen that critical loads here obtained range from 87 to 94 per cent of theoretical ones, while experimental wave numbers are equal to or slightly less than those theoretically predicted, which seem to indicate a satisfactory accuracy of the present experiments.

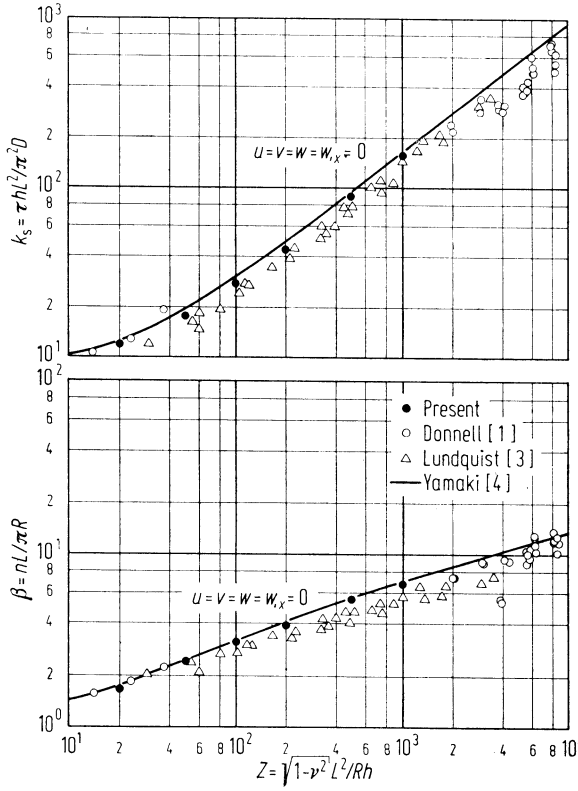


Fig. 8. Comparison between theory and experiment for the buckling of clamped cylindrical shells under torsion.

5. Summary of the Postbuckling Behavior

To clarify the influence of shell geometries on the postbuckling behavior, the torque-twist relations for shells with various values of Z are plotted together in Fig. 9. In the figure, T_{cr} is the critical torque theoretically predicted, while ψ_{cr} is the corresponding angle of twist expressed by

$$\psi_{cr} = (1 + \nu)\, LT_{cr}/\pi R^3 hE. \tag{2}$$

In Fig. 9a, thick solid lines represent the present experimental results, from which the effect of Z on the postbuckling torque-twist relation can be clearly observed. The dotted lines correspond to the experimental results obtained by Weingarten [9], which will be seen to lie considerably below the present results. Further, chain lines for $Z = 0$ represent the approximate solution for a clamped infinite strip under shear previously obtained by the author [15]. In Fig. 9b, approximate theoretical results for clamped shells by Nash [6] and those for elasti-

cally supported shells by Hayashi [7] are shown by the chain and dotted lines, respectively. It will be seen that both of these are quantitatively in poor agreement with experimental results here obtained.

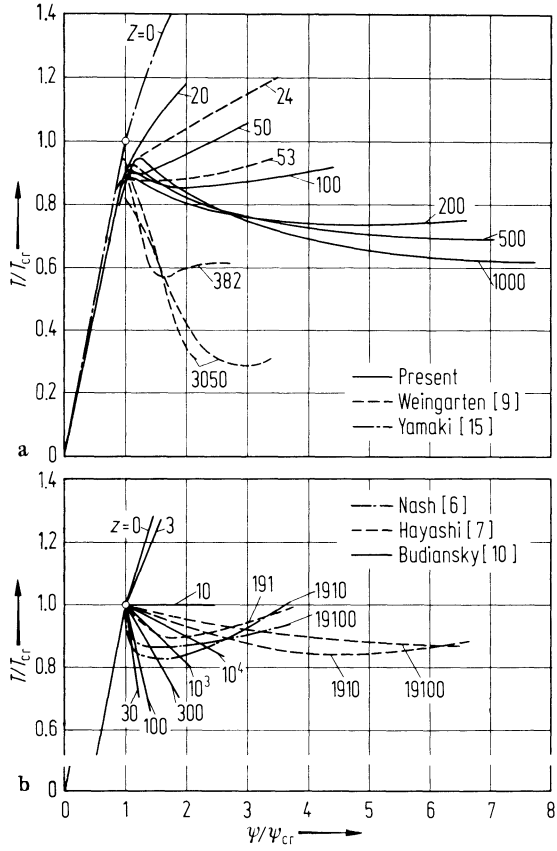


Fig. 9 a and b. Effect of Z on the postbuckling relation between T/T_{cr} and ψ/ψ_{cr}.

On the other hand, the initial postbuckling slopes determined by Budiansky [10] for completely clamped shells are depicted by solid lines, which seem to compare favourably with the present experimental results, at least for large values of Z. However, further theoretical studies will be necessary to achieve a reasonable agreement between theory and experiment.

Postbuckling relations between torque and axial shortening for each test specimen are illustrated in Fig. 10. The numeral in the parentheses denotes the number of buckled waves here observed for the special case when $R/h = 100/0.247 = 405$.

Variations of the maximum inward and outward deflections along the central sections of each cylinder, with applied torque, are shown in Fig. 11. It will be seen that with the increase in twist after buckling, the inward deflection increases monotonously, while the outward deflection soon attains its maximum and then decreases. Further, the inward deflection is much larger than the outward deflection, which exceeds ten times the shell thickness for relatively long shells with Z greater than 200.

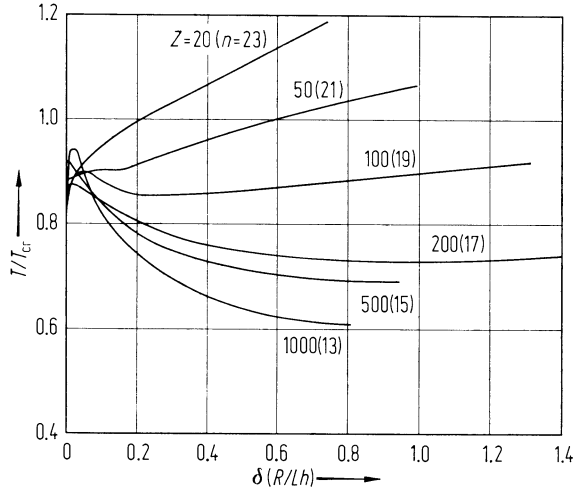


Fig. 10. Effect of Z on the postbuckling relation between T/T_{cr} and $(R/Lh)\,\delta$.

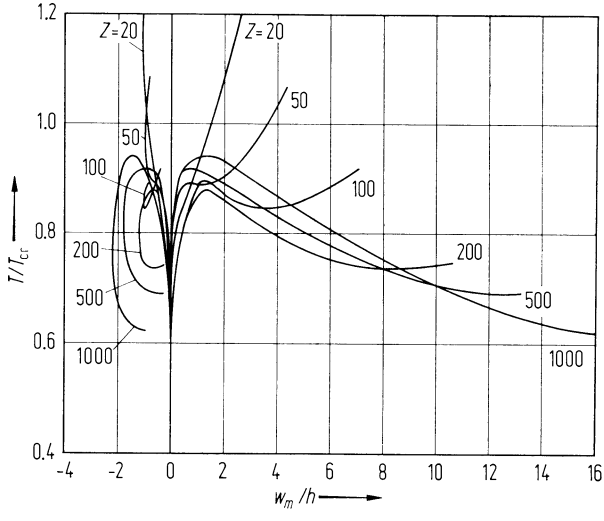


Fig. 11. Effect of Z on the postbuckling relation between T/T_{cr} and w_m/h.

It has been shown that critical loads here obtained are in fairly good agreement with those theoretically predicted. Noting that the effect of initial imperfections on the postbuckling behavior will be small in comparison with that on the critical load, the present experimental results for the postbuckling behavior will be considered to be almost equivalent to those for imperfection-free cylinders, which will be of great use for elaborate theoretical studies in the future.

Acknowledgement

The author wishes to express his thanks to Messrs. K. Otomo and T. Watanabe at the Institute of High Speed Mechanics, Tohoku University, for their assistance during the course of this work.

Appendix

For the sake of completeness, actual experimental results for the specimens with Z other than 500, which were omitted in the text, together with some pictures, are given in the following.

In each of Figs. Al through A5, the symbols a, b, c, d, and e, attached to the figures, stand for the following specifications:

a Relation between T and ψ,
b Relation between T and δ,
c Relation between T and w_m,
d Distributions of buckled waves,
e Contour lines for the buckled surface (in mm).

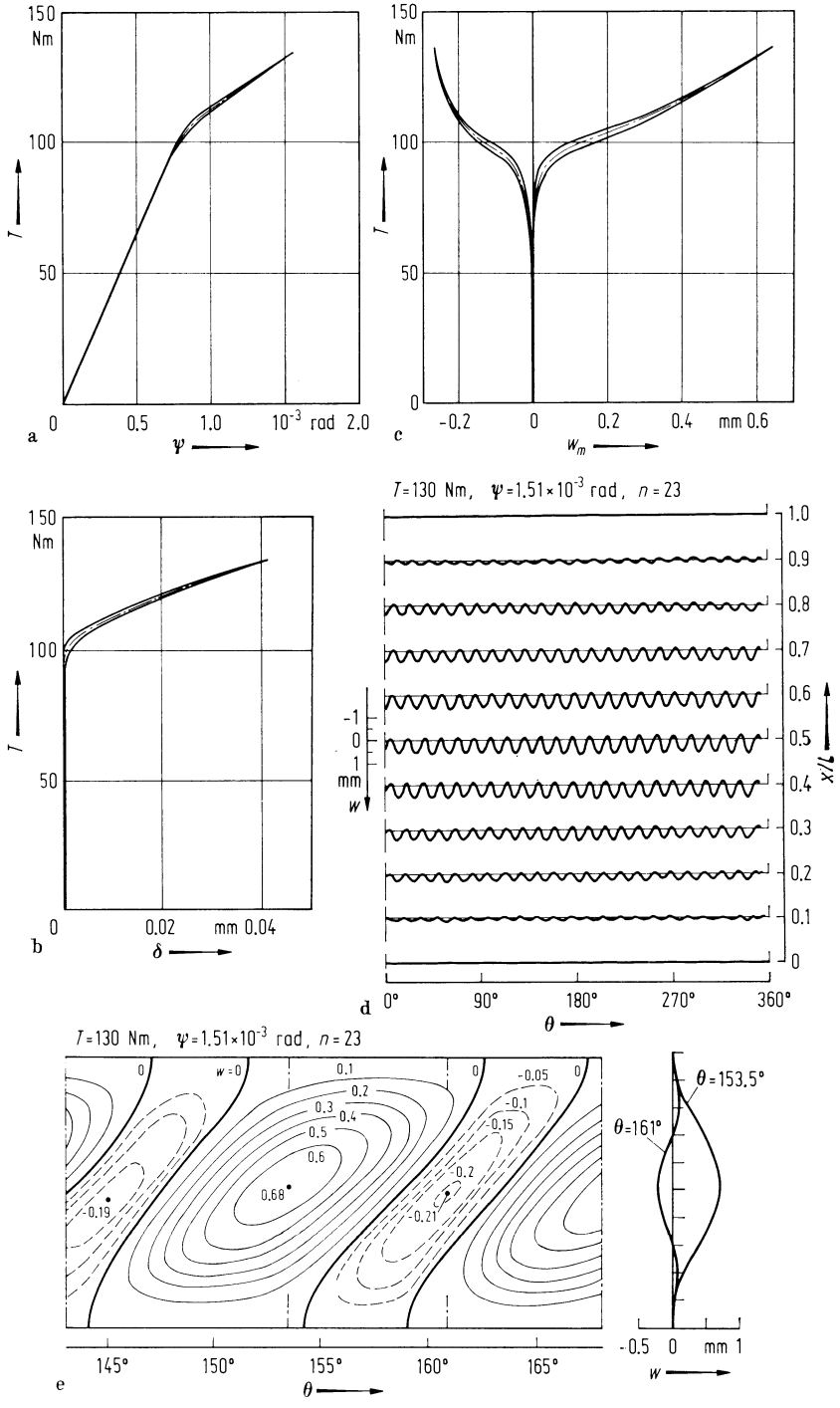


Fig. A1. Specimen No. 20-5.

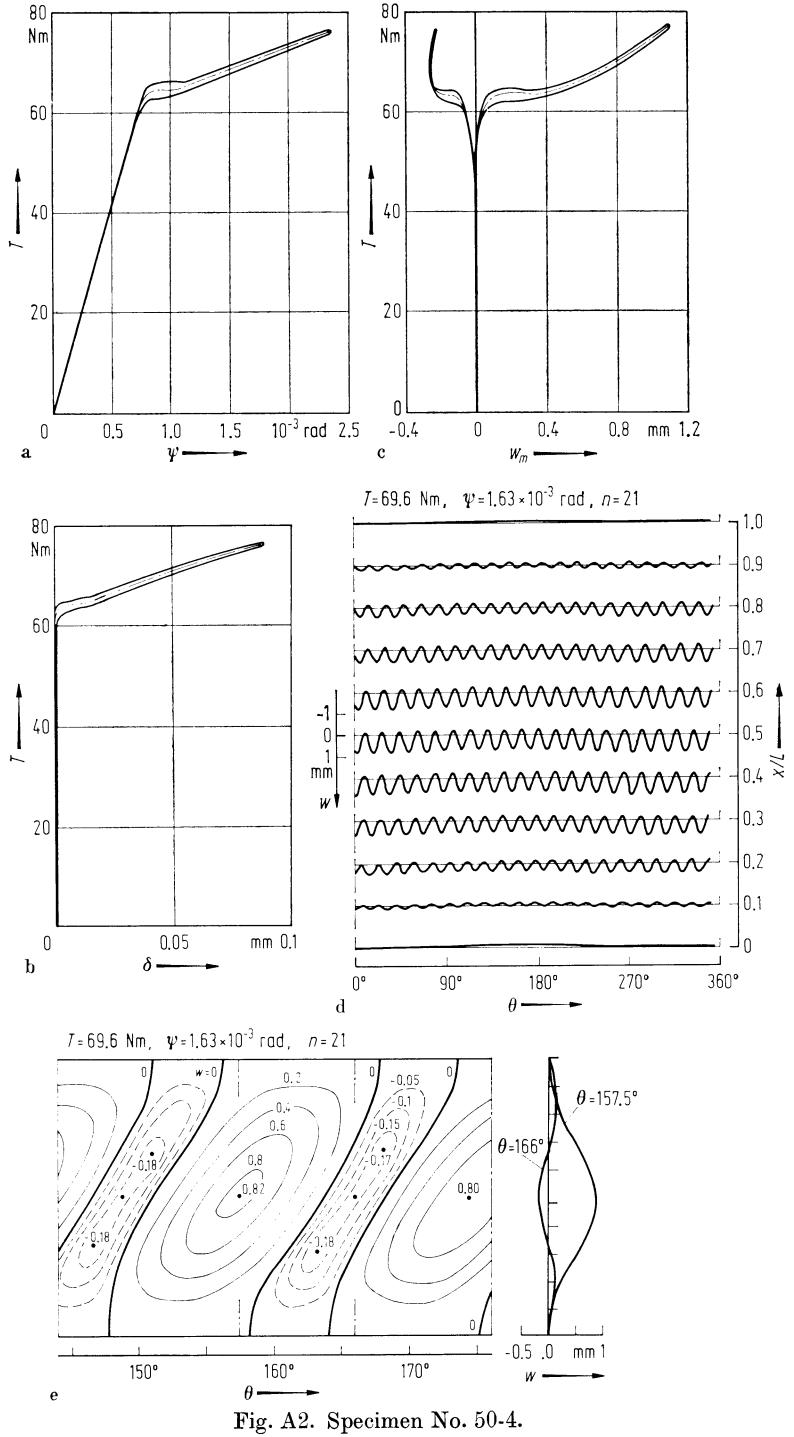


Fig. A2. Specimen No. 50-4.

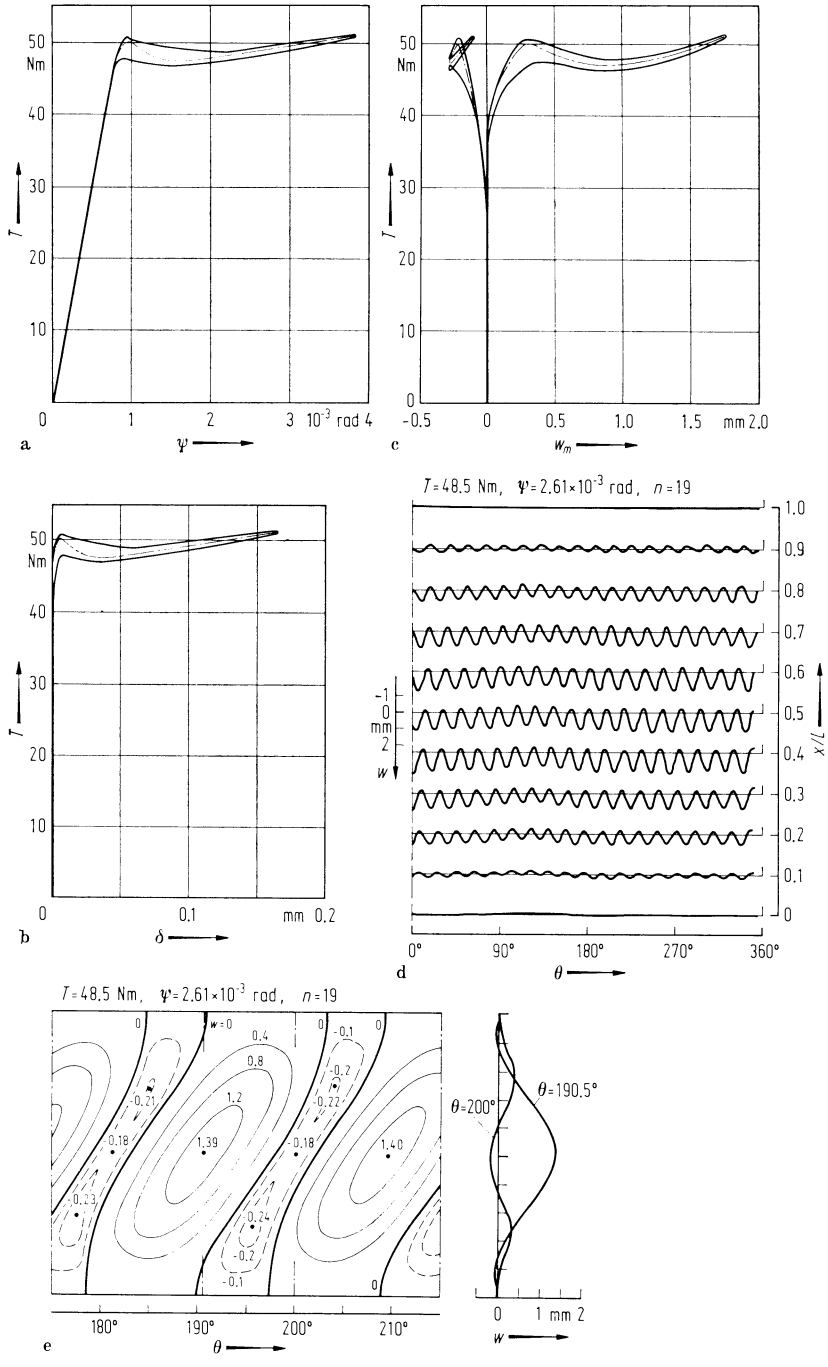


Fig. A3. Specimen No. 100-4.

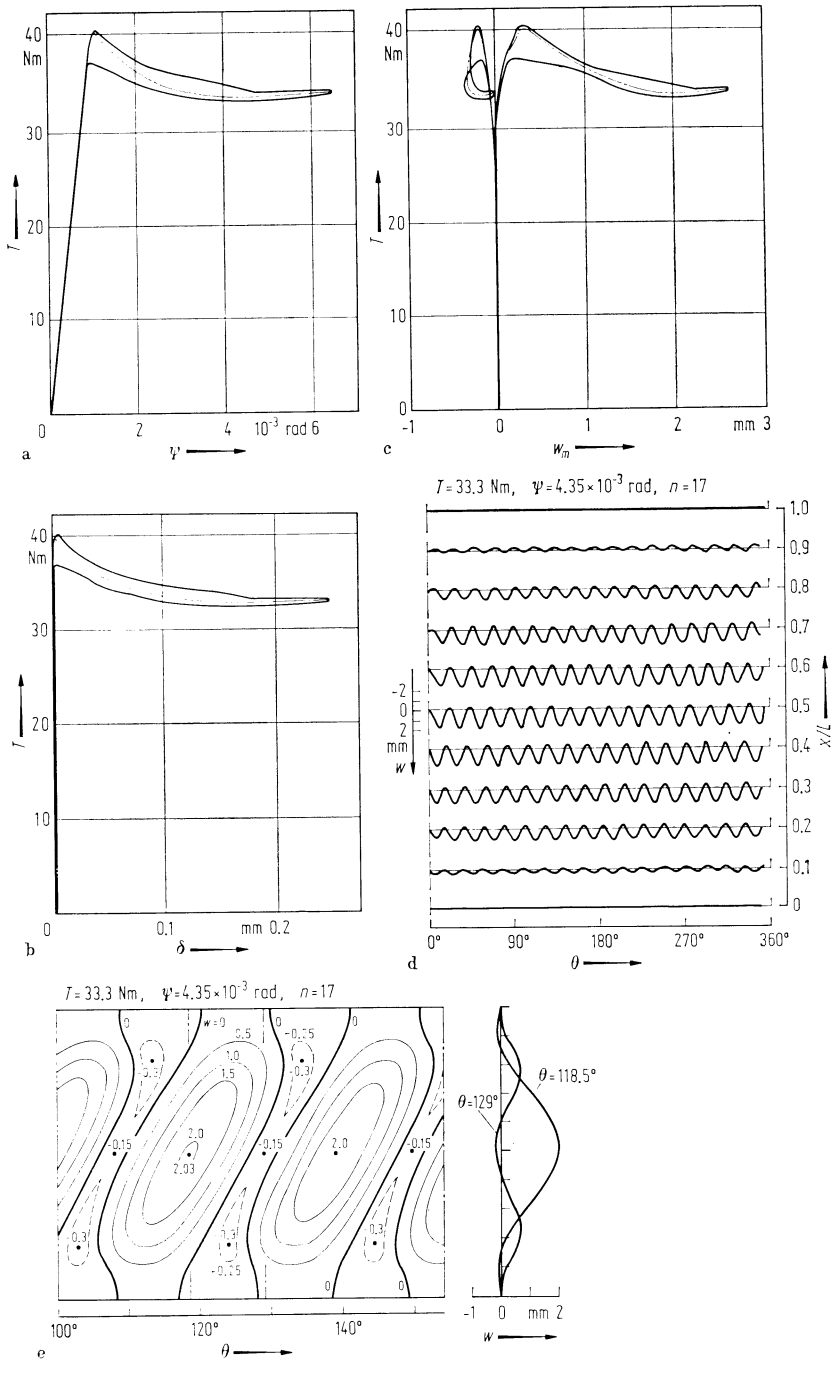


Fig. A4. Specimen No. 200-4.

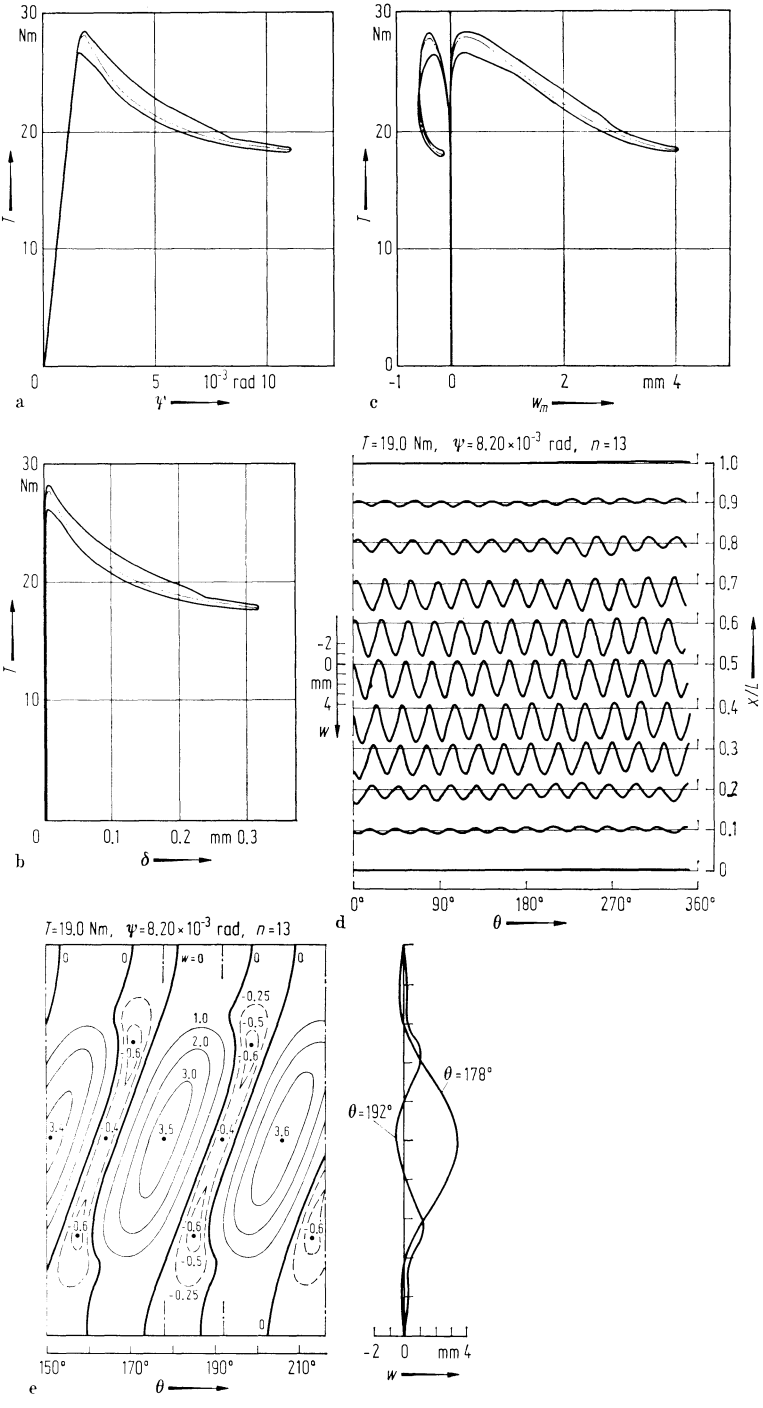


Fig. A5. Specimen No. 1000-4.

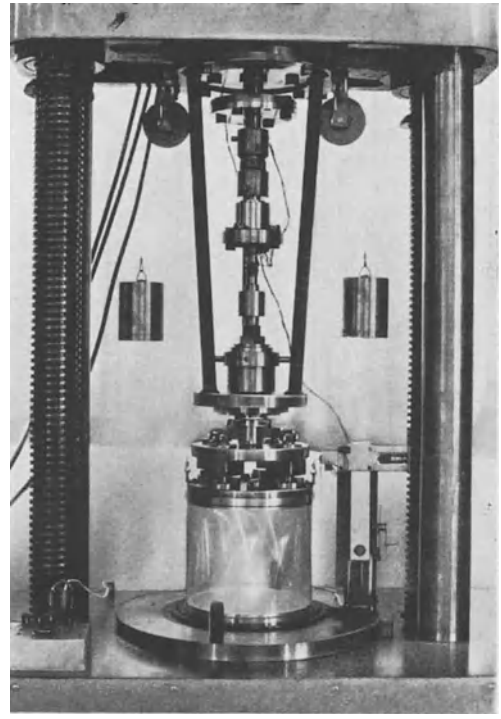


Photo 1. Test set-up.

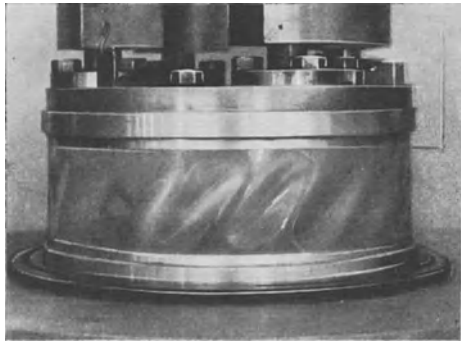


Photo 2.
Buckle pattern for specimen
No. 100-4.



Photo 3.
Buckle pattern for specimen No. 1000-4.

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